



Figure 3

Visualization of common mediator types in neural networks. **Neurons** or **attention heads** are common units of analysis. **Full layer** and **submodule** vectors are more coarse-grained, but more easily enumerable. One can also implicate a **multidimensional subspace**, which could be neuron-basis-aligned (as in a group of neurons, pictured here) or non-basis-aligned. Non-basis-aligned mediators—e.g., arbitrary **directions** in activation space—have recently become a popular mediator type due to their monosemanticity. However, discovering non-basis-aligned mediators requires external modules such as classifiers, autoencoders, or other modifications to the original computation graph. Note that while this figure depicts a Transformer, many of the mediator types generalize to other architectures (the primary exception being attention heads).

model behavior (e.g., Elhage et al. 2021; Geiger et al. 2021; Conmy et al. 2023). There have also been efforts to discover more human-interpretable mediators by moving toward latent-space structures aside from (collections of) neurons (Cunningham et al. 2024; Bricken et al. 2023; Wu et al. 2023). These methods and the units they are based on form the focus of this survey.

The lens of causal mediation analysis. For much of the history of deep learning, layers (§5.1) and neurons (§5.2) were the basic unit of study in mechanistic interpretability. They are natural units of the model (i.e., require no external modules to discover), thus making them faithful to the model’s computations by definition. In toy models, they can sometimes be human-interpretable, and their simplicity and small quantity enable researchers to exhaustively search over all of them (§6.1). Large-scale pre-trained models, however, contain far too many neurons for such methods to be tractable. Furthermore, their neurons are typically *not* straightforwardly interpretable because representations in neural networks are generally **distributed** (Hinton, McClelland, and Rumelhart 1986); in other words, there is a many-to-many relationship between neurons and concepts. Thus, the field has turned to more sophisticated abstractions like **sets of neurons**, **attention heads** (§5.2), or even **non-basis-aligned subspaces** (§5.3) that require external modules (such as probes or sparse autoencoders) to locate (§6.2). Each of these mediator types has strengths and weaknesses. A mediator type also determines, to a large extent, the kinds of concepts that can be found, and the class of methods that can be used to find them. In the following section, we more precisely define these units of analysis, and compare their strengths and weaknesses (§5). Then, we give practical context to each mediator type by describing the methods that can be used to find them (§6), and the strengths and weaknesses thereof.

Table 2

Summary of mediator types, the pros and cons of each, the search methods that are typically used to search over them, and examples of studies that employ them.

Mediator type	Strengths	Weaknesses	Common search methods	Example studies
Full layers and sub-modules (§5.1)	Small search space. Some useful applications.	Difficult to interpret. Not well-suited to explaining model behavior.	Exhaustive search (§6.1), supervised probing (§6.2.1)	Hupkes, Zuidema et al. (2017); Giulianelli et al. (2018); Hewitt and Manning (2019); Geva et al. (2023); Meng et al. (2022)
Neurons and attention heads (§5.2)	Discrete and enumerable. Relatively fine-grained; sometimes enables model control and editing.	Search space not always tractable. Often not interpretable.	Exhaustive search (§6.1), Optimization (§6.2)	Vig et al. (2020); Bau et al. (2020); Finlayson et al. (2021); Lakretz et al. (2019); Cao, Sanh, and Rush (2021)
Non-basis-aligned spaces (§5.3)	Fine-grained. Interpretable. Enables precise control of NN behaviors.	Non-enumerable. Typically requires optimization; sensitive to training setup and random variance. May not be faithful to original model.	Optimization (§6.2): supervised probing (§6.2.1), unsupervised methods such as sparse autoencoders (§6.2.2)	Wu et al. (2023); Bricken et al. (2023); Cunningham et al. (2024); Marks and Tegmark (2023); Marks et al. (2025); Ravfogel et al. (2021)

5. Mediator Types

In this section, we discuss different types of causal mediators in neural networks, and the pros and cons of each. Figure 3 visualizes a computation graph of a Transformer-based language model, and units in the graph that are often used as mediators in mechanistic interpretability studies. In mechanistic interpretability, we often do not want to treat the full computation graph as the final causal graph, as it is large and difficult to directly interpret. Thus, we typically want to build higher-level causal abstractions that capture only the most important mediators, and/or where each causal node is human-interpretable.

In this section, our primary questions are: What kinds of model components can be used as mediators? What are the strengths and weaknesses of using particular kinds of components as mediators? In this subsection, we define each mediator type. Then, in the following mediator-specific subsections, we discuss how they have been used in interpretability, and their pros and cons. Table 2 summarizes mediator types, their strengths and weaknesses, and search methods commonly used to identify each.

One possible mediator type is a **full layer**—typically the output activations or *hidden state* $\mathbf{h}^\ell$ of a specific layer ℓ (§5.1). Each index $\mathbf{h}_i^\ell$ is a **neuron** that can take some activation $\mathbf{h}_i^\ell = h_i^\ell$.¹³ One can also use the output vector of an intermediate **submodule** within the layer (e.g., an MLP), rather than the output of the whole layer. For example, in Transformers (Vaswani et al. 2017),¹⁴ a layer typically consists of two submodules: a multi-layer perceptron (MLP) and an attention block, which can be arranged either sequentially or in parallel. The outputs of these submodules $\mathbf{h}^{\ell\text{-MLP}}$ and $\mathbf{h}^{\ell\text{-Attn}}$ are also activation vectors, so we will refer to their individual dimensions as neurons as well.¹⁵

¹³ In other words, we use “neuron” to refer to any basis-aligned direction in activation space.

¹⁴ Transformers are currently the dominant architecture for language models; as such, most work in this space focuses on this architecture. However, our ideas are presented in a general way that will also apply (with minor modifications) to other neural network-based architectures, such as recurrent neural networks (Mikolov et al. 2010) and state space models (Gu, Goel, and Re 2022; Gu and Dao 2024).

¹⁵ Using the same notation emphasizes that these are mediators of the same level of granularity. However, we acknowledge that this obscures that neurons in different locations often encode different types of features.

One can also group neurons into sets, and use a neuron set as a single mediator (§5.2). A set of neurons (possibly of size 1) $\{\mathbf{h}_i^\ell, \mathbf{h}_j^\ell, \dots\}$ from a vector $\mathbf{h}^\ell$, is referred to as a **basis-aligned subspace** of $\mathbf{h}^\ell$. A one-dimensional basis-aligned subspace is equivalent to a neuron; for clarity, we will use basis-aligned subspace primarily to refer to multidimensional spaces (sets of neurons of size > 1).

Basis alignment refers to whether concept representations are aligned with specific dimensions of $\mathbf{h}^\ell$ (as contrasted with weighted combinations of dimensions). Basis alignment is a key concept: if a mediator V is aligned with the latent space basis vectors defined by a (group of) neuron(s) $\{\mathbf{h}_i^\ell, \mathbf{h}_j^\ell, \dots\}$, then we can discover it via non-parametric methods. For example, it is straightforward to exhaustively search over and intervene on individual neurons; it is less tractable, but still theoretically possible, to enumerate all $2^n - 1$ possible combinations of neurons without using any additional parameters. However, causally relevant mediators are not guaranteed to be aligned with neurons in activation space; indeed, recent work has found human-interpretable features in arbitrary directions that are *not* aligned to neuron bases (Elhage et al. 2022b; Bricken et al. 2023).

Thus, in recent studies, it is common to study **non-basis-aligned spaces** (§5.3). Each dimension in a non-basis-aligned subspace can be defined as a weighted linear combination of neuron activations. For example, to obtain a non-basis-aligned **direction**,¹⁶ we could learn coefficients α and β to weight the activations of neurons $\mathbf{h}_i^\ell$ and $\mathbf{h}_j^\ell$ (optionally with a bias term $\mathbf{b}$):

$$\mathbf{d} = \alpha \cdot \mathbf{h}_i^\ell + \beta \cdot \mathbf{h}_j^\ell + \dots + \mathbf{b}, \quad (2)$$

where neurons $\mathbf{h}_i^\ell$ and $\mathbf{h}_j^\ell$ are allowed to take their natural activations given some input x , but α and β remain fixed across inputs. Note that α and β are *not* part of the original computation graph $\mathcal{C}$. This means that discovering non-basis-aligned directions often requires external modules that weight components from the computation graph in some way—e.g., classifiers or autoencoders.

The primary trade-off between these mediator types is their granularity and quantity. This section proceeds from coarser to finer granularity and in increasing quantity. Broadly speaking, finer-grained mediators are more likely to optimize sparsity and selectivity, but are more difficult to search over, and may not be faithful to the original model if optimization is required to locate them. Coarser-grained mediators are more likely to optimize generality and are generally easier to search over, but tend to sacrifice selectivity and sparsity.

5.1 Full layers and submodules

Full layers $\mathbf{h}^\ell$ and submodules $\mathbf{h}^{\ell\text{-MLP}}$, $\mathbf{h}^{\ell\text{-Attn}}$ are relatively coarse-grained mediators; there exist only ℓ of them in a model $\mathcal{C}$.¹⁷ Early probing classifiers studied the information encoded in full layers (Shi, Padhi, and Knight 2016; Hupkes, Veldhoen, and Zuidema 2018; Belinkov et al. 2017; Conneau et al. 2018; Liu et al. 2019a; Hewitt and Manning 2019; Giulianelli et al. 2018), and recent studies that leverage classifiers as part of causal techniques still frequently do the same (e.g., Elazar et al. 2021; Marks

¹⁶ We use “direction” to refer to one-dimensional spaces.

¹⁷ Henceforth, we will refer to each of these with $\mathbf{h}^\ell$ for concision, as they are equivalent with respect to what kinds of search methods can be applied to them.